Table s1a

Multilevel Modeling Results for Smartphone Checking, Total Screen Time, Social Screen Time, Tools Screen Time in Predicting Daily Negative Affect

	Smartphone Checking		Total Screen Time		Screen Time on Social Applications		Screen Time on Tools Applications	
	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model
	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)
Fixed Effects								
Intercept, γ_{00}	**0.67 (0.08)**	**1.74 (0.49)**	**0.56 (0.09)**	**1.53 (0.47)**	**0.57 (0.08)**	**1.55 (0.47)**	**0.56 (0.04)**	**1.54 (0.45)**
Daily smartphone checking, γ_{10}	**0.00 (0.00)**	**0.00 (0.00)**	-0.00 (0.00)	-0.00 (0.00)	0.00 (0.01)	0..00 (0.01)	0.00 (0.02)	-0.00 (0.02)
Average smartphone checking, γ_{01}	-0.00 (0.00)	-0.00 (0.00)	0.01 (0.01)	0.01 (0.01)	0.01 (0.03)	0.00 (0.03)	0.08 (0.06)	0.10 (0.05)
Covariates								
Age, γ_{02}		-0.03 (0.02)		-0.03 (0.02)		-0.03 (0.02)		-0.03 (0.02)
Sex, γ_{03}		0.05 (0.07)		0.07 (0.07)		0.07 (0.07)		0.09 (0.07)
Monthly household income, γ_{04}		-0.00 (0.02)		-0.01 (0.02)		-0.00 (0.02)		-0.01 (0.02)
Subjective socioeconomic status, γ_{05}		**-0.05 (0.02)**		-0.05 (0.02)		**-0.05 (0.02)**		**-0.05 (0.02)**
Random Effects								
Intercept, μ_{0i}	0.11	0.11	0.11	0.11	0.12	0.11	0.11	0.11
Daily smartphone checking, μ_{1i}					0.00	0.00	0.01	0.01
Residual, ε_{di}	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08

Notes. Values reflect coefficient estimates in predicting daily cognitive failures. Standard errors are shown in parentheses. When the model failed to converge, random slope was removed from the model. Bolded text indicates significant coefficients (p<.05).

Table s1b

Multilevel Modeling Results for Productivity and Finance Screen Time, Entertainment and Games Screen Time, Health Screen Time and Others Screen Time in Predicting Daily Negative Affect

	Screen Time on Productivity and Finance Applications		Screen Time on Entertainment and Games Applications		Screen Time on Health Applications		Screen Time on Other Types of Applications	
	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model
	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)
Fixed Effects								
Intercept, γ_{00}	**0.67 (0.04)**	**1.65 (0.46)**	**0.60 (0.04)**	**1.58 (0.47)**	**0.59 (0.03)**	**1.59 (0.47)**	**0.58 (0.03)**	1.55 (0.46)
Daily smartphone checking, γ_{10}	0.00 (0.02)	0.00 (0.02)	-0.01 (0.01)	-0.01 (0.01)	-0.19 (0.18)	-0.19 (0.18)	-0.01 (0.02)	-0.01 (0.02)
Average smartphone checking, γ_{01}	**-0.12 (0.06)**	**-0.12 (0.06)**	-0.00 (0.02)	-0.00 (0.02)	0.53 (0.63)	0.63 90.63)	0.08 (0.07)	0.08 (0.07)
Covariates								
Age, γ_{02}		-0.03 (0.02)		-0.03 (0.02)		-0.03 (0.02)		-0.03 (0.02)
Sex, γ_{03}		0.07 (0.07)		0.07 (0.07)		0.07 (0.07)		0.08 (0.07)
Monthly household income, γ_{04}		-0.01 (0.02)		-0.01 (0.02)		-0.00 (0.02)		-0.00 (0.02)
Subjective socioeconomic status, γ_{05}		**-0.05 (0.02)**		**-0.05 (0.02)**		**-0.05 (0.02)**		-0.05 (0.02)
Random Effects								
Intercept, μ_{0i}	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11
Daily smartphone checking, μ_{1i}								
Residual, ε_{di}	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08

Notes. Values reflect coefficient estimates in predicting daily cognitive failures. Standard errors are shown in parentheses. When the model failed to converge, random slope was removed from the model. Bolded text indicates significant coefficients (p<.05).

Table s2a

Multilevel Modeling Results for Smartphone Checking, Total Screen Time, Social Screen Time, Tools Screen Time in Predicting Daily Positive Affect

	Smartphone Checking		Total Screen Time		Screen Time on Social Applications		Screen Time on Tools Applications	
	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model
	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)
Fixed Effects								
Intercept, γ_{00}	**1.59 (0.15)**	0.45 (0.96)	**2.02 (0.18)**	1.38 (0.95)	**2.15 (0.17)**	1.47 (0.95)	**2.06 (0.08)**	1.46 (0.93)
Daily smartphone checking, γ_{10}	**0.00 (0.00)**	**0.00 (0.00)**	**-0.04 (0.01)**	**-0.04 (0.01)**	-0.02 (0.02)	-0.02 (0.02)	0.04 (0.03)	0.05 (0.03)
Average smartphone checking, γ_{01}	**0.00 (0.00)**	**0.00 (0.00)**	0.00 (0.03)	0.00 (0.03)	-0.04 (0.07)	-0.03 (0.06)	-0.02 (0.12)	-0.03 (0.12)
Covariates								
Age, γ_{02}		0.03 (0.04)		0.01 (0.04)		0.01 (0.04)		0.01 (0.04)
Sex, γ_{03}		-0.13 (0.13)		-0.22 (0.13)		-0.21 (0.14)		-0.22 (0.13)
Monthly household income, γ_{04}		0.04 (0.04)		0.04 (0.04)		0.04 (0.04)		0.04 (0.04)
Subjective socioeconomic status, γ_{05}		**0.08 (0.04)**		0.08 (0.04)		0.08 (0.04)		0.08 (0.04)
Random Effects								
Intercept, μ_{0i}	0.43	0.41	0.46	0.44	0.46	0.44	0.46	0.44
Daily smartphone checking, μ_{1i}					0.01	0.01	0.00	0.00
Residual, ε_{di}	0.31	0.31	0.31	0.31	0.30	0.30	0.31	0.31

Notes. Values reflect coefficient estimates in predicting daily cognitive failures. Standard errors are shown in parentheses. When the model failed to converge, random slope was removed from the model. Bolded text indicates significant coefficients (p<.05).

Table s2b

Multilevel Modeling Results for Productivity and Finance Screen Time, Entertainment and Games Screen Time, Health Screen Time and Others Screen Time in Predicting Daily Positive Affect

	Screen Time on Productivity and Finance Applications		Screen Time on Entertainment and Games Applications		Screen Time on Health Applications		Screen Time on Other Types of Applications	
	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model
	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)
Fixed Effects								
Intercept, γ_{00}	2.07 (0.09)	1.42 (0.94)	**2.09 (0.08)**	1.48 (0.94)	**2.05 (0.62)**	1.40 (0.93)	**2.10 (0.06)**	1.44 (0.93)
Daily smartphone checking, γ_{10}	0.01 (0.03)	0.01 (0.03)	-0.02 (0.01)	-0.02 (0.01)	0.28 (0.35)	0.28 (0.35)	-0.03 (0.03)	-0.03 (0.03)
Average smartphone checking, γ_{01}	-0.04 (0.12)	-0.04 (0.12)	-0.04 (0.04)	-0.03 (0.04)	-0.25 (1.27)	-0.04 (1.26)	-0.19 (0.15)	-0.18 (0.14)
Covariates								
Age, γ_{02}		0.01 (0.04)		0.01 (0.04)		0.01 (0.04)		0.01 (0.04)
Sex, γ_{03}		-0.22 (0.13)		-0.22 (0.13)		-0.22 (1.34)		-0.21 (0.13)
Monthly household income, γ_{04}		0.04 (0.04)		0.04 (0.04)		0.04 (0.04)		0.04 (0.04)
Subjective socioeconomic status, γ_{05}		0.08 (0.04)		0.08 (0.04)		0.08 (0.04)		**0.09 (0.04)**
Random Effects								
Intercept, μ_{0i}	0.46	0.44	0.46	0.43	0.46	0.44	0.46	0.43
Daily smartphone checking, μ_{1i}								
Residual, ε_{di}	0.31	0.31	0.31	0.31	0.31	0.31	0.31	0.31

Notes. Values reflect coefficient estimates in predicting daily cognitive failures. Standard errors are shown in parentheses. When the model failed to converge, random slope was removed from the model. Bolded text indicates significant coefficients (p<.05).

Table s3

Multilevel Modeling Results for Smartphone Checking, Total Screen Time, Social Screen Time, Tools Screen Time in Predicting Daily Cognitive Failure (Distractibility)

	Smartphone Checking		Total Screen Time		Screen Time on Social Applications		Screen Time on Tools Applications	
	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model
	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)
Fixed Effects								
Intercept, γ_{00}	**0.61 (0.12)**	0.23 (0.62)	**0.59 (0.14)**	0.31 (0.61)	**0.70 (0.13)**	0.47 (0.59)	**0.67 (0.06)**	0.40 (0.60
Daily smartphone checking, γ_{10}	**0.00 (0.00)**	0.00 (0.00)	**-0.02 (0.01)**	**-0.02 (0.01)**	**-0.05 (0.02)**	**-0.04 (0.01)**	**-0.07 (0.03)**	**-0.07 (0.03)**
Average smartphone checking, γ_{01}	0.00 (0.00)	0.00 (0.00)	0.01 (0.02)	0.01 (0.02)	-0.03 (0.05)	-0.04 (0.04)	-0.08 (0.09)	-0.07 (0.07)
Covariates								
Age, γ_{02}		0.00 (0.02)		-0.00 (0.02)		-0.00 (0.02)		-0.00 (0.02)
Sex, γ_{03}		-0.08 (0.09)		-0.10 (0.08)		-0.10 (0.08)		-0.09 (0.08)
Monthly household income, γ_{04}		0.01 (0.02)		0.01 (0.02)		0.01 (0.02)		0.00 (0.02)
Subjective socioeconomic status, γ_{05}		-0.00 (0.03)		-0.01 (0.03)		-0.01 (0.03)		-0.01 (0.03)
Daily stressor exposure, B_2		**0.17 (0.04)**		**0.17 (0.04)**		**0.16 (0.04)**		**0.18 (0.04)**
Average stressor exposure, γ_{06}		**0.57 (0.14)**		**0.59 (0.14)**		**0.61 (0.13)**		**0.54 (0.14)**
Daily positive affect, B_3		0.01 (0.03)		0.02 (0.03)		0.01 (0.03)		0.02 (0.03)
Average positive affect, γ_{07}		-0.08 (0.05)		-0.07 (0.05)		-0.06 (0.05)		-0.07 (0.05)
Daily negative affect, B_4		**0.40 (0.05)**		**0.40 (0.05)**		**0.41 (0.05)**		**0.40 (0.05)**
Average negative affect, γ_{08}		**0.54 (0.10)**		**0.54 (0.11)**		**0.53 (0.10)**		**0.55 (0.11)**
Random Effects								
Intercept, μ_{0i}	0.27	0.15	0.27	0.15	0.27	0.15	0.27	0.15
Daily smartphone checking, μ_{1i}					0.00	0.00	0.00	0.00
Residual, ε_{di}	0.21	0.19	0.21	0.19	0.21	0.18	0.21	0.19

Notes. Values reflect coefficient estimates in predicting daily cognitive failures. Standard errors are shown in parentheses. When the model failed to converge, random slope was removed from the model. Bolded text indicates significant coefficients (p<.05).

Table s4

Multilevel Modeling Results for Smartphone Checking, Total Screen Time, Social Screen Time, Tools Screen Time in Predicting Daily Cognitive Failure (Blunders)

	Smartphone Checking		Total Screen Time		Screen Time on Social Applications		Screen Time on Tools Applications	
	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model
	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)
Fixed Effects								
Intercept, γ_{00}	**0.18 (0.07)**	-0.18 (0.41)	0.15 (0.08)	-0.16 (0.40)	**0.21 (0.08)**	-0.18 (0.04)	**0.23 (0.04)**	-0.08 (0.39)
Daily smartphone checking, γ_{10}	**0.00 (0.00)**	**0.00 (0.00)**	-0.01 (0.01)	-0.00 (0.01)	-0.02 (0.01)	-0.01 (0.01)	-0.03 (0.02)	-0.03 (0.02)
Average smartphone checking, γ_{01}	0.00 (0.00)	0.00 (0.00)	0.01 (0.01)	0.01 (0.01)	0.01 (0.03)	0.02 (0.02)	0.02 (0.06)	-0.02 (0.05)
Covariates								
Age, γ_{02}		-0.00 (0.01)		-0.00 (0.01)		0.00 (0.01)		-0.00 (0.01)
Sex, γ_{03}		**-0.14 (0.06)**		**-0.15 (0.06)**		**-0.14 (0.05)**		**-0.15 (0.06)**
Monthly household income, γ_{04}		**0.03 (0.01)**		**0.03 (0.01)**		0.02 (0.01)		0.03 (0.02)
Subjective socioeconomic status, γ_{05}		-0.00 (0.02)		-0.00 (0.02)		-0.01 (0.02)		-0.00 (0.02)
Daily stressor exposure, B_2		0.03 (0.03)		0.03 (0.03)		0.03 (0.02)		0.03 (0.02)
Average stressor exposure, γ_{06}		0.09 (0.09)		0.11 (0.09)		0.11 (0.09)		0.09 (0.09)
Daily positive affect, B_3		**0.06 (0.02)**		**0.06 (0.02)**		**0.06 (0.02)**		**0.06 (0.02)**
Average positive affect, γ_{07}		0.03 (0.03)		0.04 (0.03)		0.05 (0.03)		0.04 (0.03)
Daily negative affect, B_4		**0.17 (0.04)**		**0.18 (0.04)**		**0.17 (0.03)**		**0.17 (0.04)**
Average negative affect, γ_{08}		**0.53 (0.07)**		**0.52 (0.07)**		**0.50 (0.07)**		**0.53 (0.07)**
Random Effects								
Intercept, μ_{0i}	0.10	0.06	0.10	0.06	0.10	0.06	0.10	0.06
Daily smartphone checking, μ_{1i}					0.00	0.00	0.01	0.01
Residual, ε_{di}	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.09

Notes. Values reflect coefficient estimates in predicting daily cognitive failures. Standard errors are shown in parentheses. When the model failed to converge, random slope was removed from the model. Bolded text indicates significant coefficients (p<.05).

Table s5

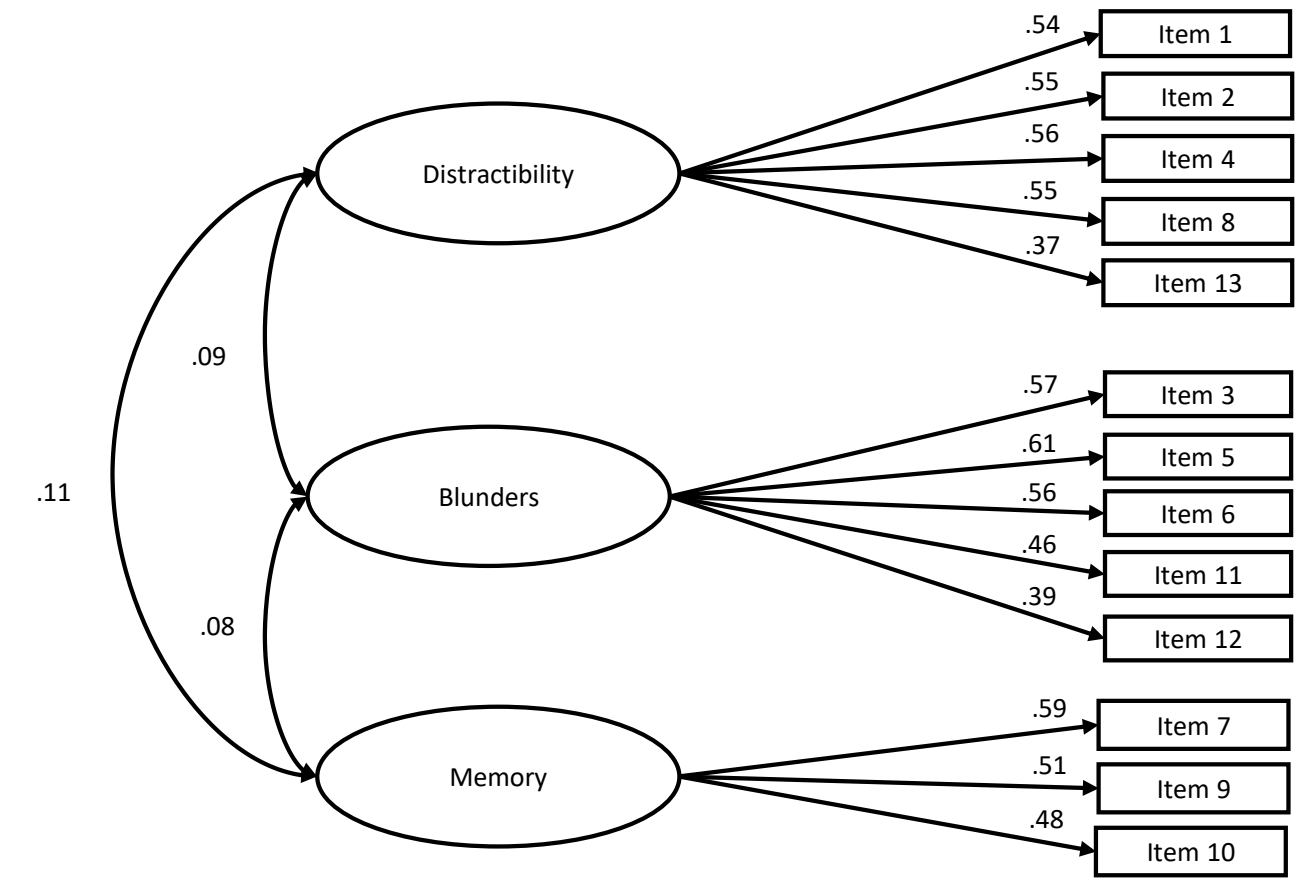
Multilevel Modeling Results for Smartphone Checking, Total Screen Time, Social Screen Time, Tools Screen Time in Predicting Daily Cognitive Failure (Memory Failure)

	Smartphone Checking		Total Screen Time		Screen Time on Social Applications		Screen Time on Tools Applications	
	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model	Unadjusted Model	Adjusted Model
	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)	Coeff. (*SE*)
Fixed Effects								
Intercept, γ_{00}	**0.34 (0.10)**	0.21 (0.55)	0.28 (0.11)	0.18 (0.54)	**0.43 (0.10)**	0.33 (0.51)	**0.38 (0.05)**	0.29 (0.53)
Daily smartphone checking, γ_{10}	0.00 (0.00)	0.00 (0.00)	-0.01 (0.01)	-0.01 (0.01)	**-0.03 (0.01)**	**-0.03 (0.01)**	-0.04 (0.03)	-0.04 (0.03)
Average smartphone checking, γ_{01}	0.00 (0.00)	0.00 (0.00	0.01 (0.02)	0.02 (0.01)	-0.02 (0.04)	-0.02 (0.03)	-0.02 (0.08)	-0.03 (0.06)
Covariates								
Age, γ_{02}		-0.01 (0.02)		-0.01 (0.02)		-0.01 (0.02)		-0.01 (0.02)
Sex, γ_{03}		-0.15 (0.08)		**-0.16 (0.07)**		**-0.17 (0.07)**		**-0.16 (0.07)**
Monthly household income, γ_{04}		0.03 (0.02)		0.03 (0.02)		0.03 (0.02)		0.03 (0.02)
Subjective socioeconomic status, γ_{05}		-0.01 (0.02)		-0.01 (0.02)		-0.01 (0.02)		-0.01 (0.02)
Daily stressor exposure, B_2		**0.09 (0.03)**		**0.09 (0.03)**		**0.08 (0.03)**		**0.09 (0.03)**
Average stressor exposure, γ_{06}		**0.33 (0.12)**		**0.35 (0.12)**		**0.38 (0.12)**		**0.33 (0.12)**
Daily positive affect, B_3		0.04 (0.02)		0.04 (0.02)		0.04 (0.02)		0.04 (0.02)
Average positive affect, γ_{07}		-0.01 (0.03)		-0.01 (0.04)		0.02 (0.04)		-0.01 (0.04)
Daily negative affect, B_4		**0.20 (0.05)**		**0.21 (0.05)**		**0.20 (0.05)**		**0.21 (0.05)**
Average negative affect, γ_{08}		**0.54 (0.09)**		**0.53 (0.09)**		**0.51 (0.09)**		**0.54 (0.09)**
Random Effects								
Intercept, μ_{0i}	0.18	0.12	0.18	0.12	0.18	0.12	0.18	0.12
Daily smartphone checking, μ_{1i}					0.00	0.00	0.02	0.01
Residual, ε_{di}	0.17	0.16	0.17	0.16	0.16	0.16	0.16	0.16

Notes. Values reflect coefficient estimates in predicting daily cognitive failures. Standard errors are shown in parentheses. When the model failed to converge, random slope was removed from the model. Bolded text indicates significant coefficients (p<.05).

Figure s1

Multilevel Factor Model of Cognitive Failure in Everyday Life Scale



Note: Latent variables are depicted by ellipses while manifest variables are depicted by boxes. Standardized factor loadings are depicted by single-headed arrows and inter-factor correlations are depicted by double-headed arrows. All loadings and correlations were statistically significant at $p<.05$